

Issues and Challenges for Teaching Successful Online Courses in Higher Education: A Literature Review

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Abstract

Online education changes all components of teaching and learning in higher education. Many empirical studies have been conducted to examine issues in delivering online courses; however, few have synthesized prior studies and provided an overview on issues in online courses. A review of literature using Cooper's framework was conducted to identify such issues. Three major categories of findings were identified: issues related to online learners, instructors, and content development. Learners' issues included learners' expectations, readiness, identity, and participation in online courses. Instructors' issues included changing faculty roles, transitioning from face-to-face to online, time management, and teaching styles. Content issues included the role of instructors in content development, integration of multimedia in content, role of instructional strategies in content development, and considerations for content development. To address these challenges in online education, higher education institutions need to provide professional development for instructors, trainings for learners, and technical support for content development.

Keywords

distance education and telelearning, teaching or learning strategies, postsecondary education, pedagogical issues, human–computer interface

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Introduction

Online education has become increasingly popular in the U.S. higher education within the last two decades, and most higher education institutions believe that this method of instruction will be critical for the future of higher education (Allen & Seaman, 2014). The accessibility of the internet and flexibility of online courses have made online education an integral part of higher education (Li & Irby, 2008; Luyt, 2013; Lyons, 2004). In addition, financial issues facing many higher education institutions and students' demands shift the focus of these institutions more toward using online education (Limperos, Buckner, Kaufmann, & Frisby, 2015). Given the opportunities that online education provides for faculty, students, and institutions, the amount of attention it has received is not surprising (Konetes, 2011).

Many empirical studies have been conducted to examine the quality of online courses from various aspects. Studies have identified and examined critical issues affecting quality of online education such as communication, technology, time management, pedagogy, and assessment (Bassoppo-Moyo, 2006; Conaway, Eston, & Schmit, 2005; Ko & Rossen, 2010; Limperos, et al., 2015). There are also organizations such as Quality Matters and Online Learning Consortium that focus on improving quality of online education in higher education by providing resources as well as opportunities for collaboration on curriculum development. However, the literature pertaining to online education needs literature reviews that further synthesize and integrate the empirical studies' results and provide an integrative report on existing challenges in teaching online courses. Often online educators must go through the daunting task of sifting through the increasingly expanding literature to identify these issues for themselves (Mayes, Luebeck, Yu Ku, Akarasriworn, & Korkmaz, 2011).

Furthermore, because of continued reports of high dropout rates and achievement problems in online courses (Luyt, 2013; Morris, Xu, & Finnegan, 2005; Tyler-Smith, 2006), conducting such an investigation and providing the results increasingly become critical in order to inform educators about considerations and changes necessary for improving the quality of online courses. The purpose of this study was, therefore, to inform educators about the major issues and strategies that affect the quality of teaching online courses in higher education. We have examined the literature to identify major challenges and issues in teaching online higher education courses, organized and provided the issues under topical classification, and provided some suggestions to address the issues for online educators.

Method

To achieve the purpose of the study, a literature review was conducted using Cooper's (1988) procedure for synthesizing literature to (a) formulate the problem, (b) collect data, (c) evaluate the appropriateness of the data, (d) analyze and interpret relevant data, and (e) organize and present the results. The results were

then depicted in a model which shows the issues affecting the teaching of online courses and the relationship among these issues. For the purpose of this study, online education is defined as postsecondary and credit bearing coursework completely delivered through online courses via a learning management system (LMS) such as Blackboard or Moodle. The focus of this study is on online courses offered via a LMS by academic higher education institutions including public, private, and for-profit universities in the United States. Students enroll in online courses as part of their degree requirements.

Problem Formulation

The problem is that the rapid integration of online education into higher education has diverted educators' attention from closely identifying major challenges in teaching online courses and forming a combined overview based on previous studies. Studies have suggested various issues, but the issues were not categorized and combined under any topical classification to provide online educators with an organized overview of the issues (Mayes et al., 2011). For example, Brooks (2003) referred to attitudes of instructors as a major issue affecting teaching online courses. Arbaugh (2005) considered technology, behavioral characteristics of the learners, and instructors' teaching style as essential challenges. Jacobs (2014) suggested assessment of students is a major issue in online courses. Yueng (2001) referred to instructor and student support, course development, course structure, and how the institution evaluates online learning as major issues.

Such variety in discrete reporting may cause educators to spend additional time reviewing the studies to capture an entire view about issues in teaching online courses. To help address the problem and form a combined classification of reported issues for teaching online courses, the following question guided this review: What are the major categories of issues and challenges that affect teaching online courses in higher education institutions in the United States?

Data Collection

The purpose of data collection was to find empirical studies including quantitative, qualitative, mixed methods, and literature reviews conducted on identifying challenges in online education published in peer-reviewed journals within the range of 1990 to 2015. The keywords that were used included "online courses and issues," "online education and challenges," "web-based instruction," "online teaching and issues," "course redesign," "instructional design and online learning," "curriculum design and online faculty," "distance learning," "e-learning," and "online instruction and issues." The databases that were used for literature research included Google Scholar, Educational Resources Information Center (ERIC), JSTOR, Teaching and Learning Journals, Conference Proceedings, and EBSCO HOST.

Data Evaluation and Analysis

Based on the described procedure, 104 articles were found. Content analysis approach (Strauss, 1987) was used to analyze the collected articles based on the main topic of the studies. Online course disciplines, demographics of students, and level of courses (i.e., undergraduate vs. graduate) were not considered as factors for exclusion or inclusion of the studies. The focus was on including studies not on an institutional level in higher education but on the level of individual online courses within the institution. In other words, studies that were included in the review examined teaching and learning issues within online courses, not institutional issues related to online courses such as administration policies, budgets, or online program development. The collected articles were categorized into major topical themes. New themes were added until the data themes reached saturation, meaning that all new data could be categorized under the already developed themes. Of the 104 articles that were found on online courses, 25 on learners' issues, 23 focused on content design, 45 focused on issues related to instructors' issues, 11 focused on mixed issues of instructors, learners, and content. The remaining articles that did not discuss specific issues in the above three aspects were excluded from this review.

Limitation

The literature search for the studies was extensive and systematic using a framework to find as many related studies as possible; however, the review was not exhaustive. The findings were limited to the results that emerged in the searched databases using the aforementioned keywords. It is reasonable to assume there were other related studies in the literature that did not emerge and were not included in this review.

Results

As a result of reviewing the studies and categorizing them into major topical themes, the following three major themes of issues related to learners, instructors, and content emerged.

Issues Related to Learners

The review of literature revealed that issues related to learners may be summarized into learners' expectations, readiness, identity, and participation in online courses as shown in Table 1.

Learners' expectations. Learners' expectations can be challenging and can also interfere with effectively teaching online courses (Li & Irby, 2008; Luyt, 2013). Some learners may have inappropriate expectations such as expecting instant

Table 1. Learners’ Issues and Related Sources.

Learners’ issues	Sources
Expectations	• Li and Irby (2008) • Lyons (2004)
Readiness	• Aragon, Johnson, and Shaik (2002)
• Learning style • Cultural differences • Technical skills	• Hill (2002) • Hung, Chou, Chen, and Own (2010) • Luyt (2013) • Mayes et al. (2011) • Peng, Tsai, and Wu (2006) • Roper (2007) • Smith et al. (2003) • Tsai and Lin (2004)
Identity	• Goodyear and Zenios (2007)
• Isolation	• Koole (2014) • Lapadat (2007) • McInnery and Roberts (2004)
Participation	• An and Frick (2006) • Ching, and Hsu (2015) • Hew and Hara (2007) • Hrastinski (2008, 2009) • Ice, Curtis, Phillips, and Wells (2007) • Morris et al. (2005) • Olesova, Richardson, Weasenforth, and Meloni, (2011) • Romiszowski and Mason (2004) • Vonderwell and Zachariah (2005) • Wise, Speer, Marbouti, and Hsiao (2013)

feedback on their online comments and assignments or may appear rude and demanding in their emails. Some may question their grades and others may not take the assignment deadlines seriously (Li & Irby, 2008; Lyons, 2004). Instructors are suggested to minimize these inappropriate expectations by clearly communicating their course rules and policies at the beginning of the course.

Learners’ readiness. Learners’ readiness to attend online courses is one of the major issues discussed in literature (Hung, Chou, Chen, & Own, 2010; Smith, Murphy, & Mahoney, 2003). Not all learners can successfully participate in online courses. Identifying and adopting learning styles and skills required to participate in online courses can be challenging for learners (Mayes et al., 2011; Luyt, 2013). Mostly the learners need to be self-motivated and self-directed. Online instructors should be ready to help learners who lack the required

learning skills. To help learners, the major aspects or dimensions of readiness should be further clarified. Literature indicated that learners' technical skills related to use of computers and the Internet (Peng, Tsai, & Wu, 2006), their perceptions and attitudes toward the Internet (Tsai & Lin, 2004), their cultural and non-English backgrounds (Luyt, 2013), and their time management skills (Hill, 2002; Roper, 2007) are considered important for shaping learners' readiness to participate in online courses.

To help learners identify the required skills, a useful overarching model with five major readiness dimensions along with an instrument to measure the dimensions was suggested by Hung et al. (2010). The five dimensions include self-directed learning, motivation for learning, computer and Internet self-efficacy, online communication self-efficacy, and learner control (Hung et al. 2010). First, self-directed learning is defined as a process in which learners take responsibility in understanding their learning needs, establishing their learning goals, and implementing learning strategies and evaluation (Knowles, 1975). Online courses provide a great extent of flexibility and autonomy for the learners. Learners with higher levels of self-directed learning are more successful in online settings (Lin & Hsieh, 2001). Second, motivation refers to the "need to do something out of curiosity and enjoyment" (Hung et al., 2010, p. 1082). Motivation for learning in online settings plays a critical role in the success of learners, aligning learners' efforts with learners' desires, and increasing learner retention (Saadé, He, & Kira, 2007). Third, computer and Internet self-efficacy refers to learners' perceptions about their skills to use computers and the Internet to accomplish a task. Researchers noted that this efficacy is related to complex tasks such as trouble shooting problems. Learners with higher self-efficacy perform better in online courses (Tsai & Tsai, 2003). Fourth, online communication self-efficacy refers to learners' perceptions about their abilities to communicate in online settings. Learners with higher communication self-efficacy were reported to better perform in online settings (McVay, 2000; Roper, 2007). Finally, learner control refers to the degree which learners can direct their learning experiences (Hung et al., 2010; Shyu & Brown, 1992). Online settings as compared with traditional face-to-face settings offer learners with more control over their learning materials. Online learners may choose the sequence, pace, and amount of content and may follow a more individualized approach. Learners perform better when they are provided with more control (Hung et al., 2010, Reigeluth & Stein, 1983; Wang & Beasley, 2002). Instructors may use the Hung et al. (2010) validated instrument to identify online learners' readiness levels within these five dimensions and support the learners who are not well-prepared to participate in online courses.

Learners' identity. Learners may feel isolated and disconnected in online courses (McInnery & Roberts, 2004), which may affect learning. Affiliation with the community of learning influences learners' sense of identity and learning

(Koole, 2014). Particularly from a social constructionist perspective, learners and the community within which they interact socially cocreate their identities. Therefore, it is critical to help learners develop a shared sense of belonging, purpose, and norms (Koole, 2014; Lapadat, 2007). A strong sense of identity along with belonging to the knowledge community as a valued member plays a critical role in effective knowledge building (Goodyear & Zenios, 2007). To help learners achieve a sense of identity, Koole (2014) developed a Web of Identity model with five components of technical, political, structural, cultural, and personal dramaturgical or performance to succeed in online settings. Technical dramaturgical encompasses asynchronous time to think and semi-permanent time to review. Political dramaturgical includes sharing power, persuading, dominating, and controlling. Structural dramaturgical includes status seeking, including or excluding, role taking, and realigning structure. Cultural dramaturgical includes behavioral expectations, sharing purpose, and using symbols. Personal-agency dramaturgical includes interaction styles and sharing personal information.

Learners' participation. Learners' nature of participation and engagement in online settings is another major issue discussed by researchers. Some researchers considered participation through interacting with peers and instructors by writing (Romiszowski & Mason, 2004; Vonderwell & Zachariah, 2005), while others suggested that learners who observed the interaction and spent time on reading more than writing were still engaged in learning (Hrastinski, 2008, 2009; Wise, Speer, Marbouti, & Hsiao, 2013). Online listening or observing is a complex phenomenon and a substantial component of learners' participation in online discussion. A group of researchers in their study of 96 participants in 3-week online undergraduate courses found that learners spent three quarters of their time listening or observing in online discussions (Wise et al., 2013). Romiszowski and Mason (2004) argued the assumption that infrequent contributors are "passive recipients rather than actively engaged in learning" (p. 399) has been rarely challenged in research. However, online listening or observing is a form of active learning, as it consists of engagement with the content, thought, and reflection (Hrastinski, 2009). This is grounded in the concept of vicarious learning where learning occurs by observing others' active dialogue (Kolb, 1984; McLendree, Stenning, Mayes, Lee, & Cox, 1998). The implication for online instructors is to recognize and support the nature of learners' online participation. It is an oversimplified approach to judge learners' participation only by the quantity or length of their online postings. A more inclusive framework based on the social perspectives on learning discussed by Vygotsky (1978), Wenger (1998), and Säljö (2000), through which participation includes doing, talking, thinking, and feeling, which occur in both online and offline environments (Hrastinski, 2008, 2009; Wise et al., 2013), may be used to more appropriately judge learners' online participation. For example, Morris, Finnegan, and Sz-Shyans (2005) used both frequency variables (e.g., counting postings) and duration variables (e.g.,

seconds spending viewing postings or content pages and number of postings viewed) to identify predictor variables for learners' final course grades in their study of 354 online learners at the University of Georgia and found the duration variables were the predictors of the final grades.

Additionally, the literature indicates that learners' participation in online discussions can be enhanced by mixing audio or video discussion with online text discussions (An & Frick, 2006; Ching & Hsu, 2015; Hara & Hew, 2007; Ice, Curtis, Phillips, & Wells, 2007; Olesova, Richardson, Weasenforth, & Meloni, 2011). Audio or video components enhance communication and connection with peers, encourage learners to participate in the discussions, and support learners to elaborate on their responses, as it facilitates less time consuming communication as compared with text discussion (An & Frick, 2006; Ching & Hsu, 2015; Hara & Hew, 2007). One drawback of using audio or video discussion was the difficulty of extracting ideas from long audio or video comments as compared with text comments (Ching & Hsu, 2015). In general, audio or video discussion has been recommended to be used for community building, ice breaking activities, and supplementary for clarification and elaboration on text postings.

Issues Related to Content

The review of literature revealed that issues related to content may be summarized into the role of instructors in content development, integration of multimedia in content, role of instructional strategies in content development, and considerations for content development as shown in Table 2.

Content development and instructors. In a majority of online courses, instructors teach with predefined content. In these courses, instructors face the issue of lack of empowerment (Evrin, Correia, & Thompson, 2011). In such situations, the role of instructors in creating, shaping, and integrating their own experiences into the content of the courses has been downplayed (Rennert-Ariev, 2008). Online instructors are encouraged to engage in designing the content and adopting an autonomous and active role through constantly criticizing their assumptions toward online teaching (Evrin et al., 2011).

Instructors may be responsible for preparing and planning materials for online courses. The task of generating new materials or adjusting the materials from face-to-face classes to an online setting can be very challenging (Li & Irby, 2008). Often proper training and support has not been provided to instructors who are transitioning course content from face-to-face to online settings (Kyei-Blankson & Keengwe, 2011).

Content cannot simply be copied from a face-to-face to an online setting. Koehler, Mishra, Hershey, and Peruski (2004) encouraged instructors to take content, pedagogy, and technology into account when designing online courses. However, instructors may not be willing to change their teaching strategies when

Table 2. Content Issues and Related Sources.

Content issues	Sources
Development and instructors	• Allen and Seaman (2010, 2014) • Evrim, Correia and Thompson (2011) • Hoyt and Oviatt (2013) • Herman (2013) • Kyei-Blankson and Keengwe (2011) • Koehler, Mishra, Hershey, and Peruski (2004) • Li and Irby (2008) • Parker (2003) • Rennert-Ariev (2008)
Content and multimedia	• Almala (2005) • Hathaway (2013) • Mayer (2014) • Miller (2014) • Niess and Gillow-Wiles (2013) • Yue et al. (2013)
Content and instructional strategies	• Chametzky (2014) • Jacobs (2014) • Luyt (2013) • Niess and Gillow-Wiles (2013) • Twigg (2005)
Content and consideration	• Allen et al. (2013) • Gikandi, Morrow, and Davis (2011) • Miller (2014)

transitioning from a face-to-face to an online course (Barrett, 2010). In addition to a lack of training and support, instructors also perceive a lack of incentives in designing and delivering online courses (Allen & Seaman, 2010). Offering appropriate incentives increases an instructor's willingness to design and deliver an online course (Hoyt & Oviatt, 2013). However, institutions are encouraged to communicate with instructors to determine what incentives to offer as extrinsic monetary incentives are not necessarily what faculty desire (Herman, 2013). Intrinsic motivators that may positively influence instructors to deliver an online course are a flexible schedule and their own self-satisfaction. Other extrinsic motivators that may be more influential than monetary incentives include a decreased workload and release time (Parker, 2003).

Content and multimedia. Content may be developed based on strategies such as integrating multimedia to enhance the learning experience using constructivist theory principles (Almala, 2005). Examples of multimedia include learning

games, videos, and simulations. It is important to note that simply incorporating multimedia into the design of online courses is not always the right answer. Instructors need to ask themselves what the technology will add to the learning activity. Multimedia used in the wrong way can be a detriment to the learning process (Yue, Bjork, & Bjork, 2013). According to multimedia learning theory, there are three instructional design approaches to enhance learning: less-is-more, more-is-more, and focused-more-is-more (Mayer, 2014). The less-is-more approach focuses on reducing or eliminating extraneous material to avoid any distraction in the learning process. The more-is-more approach focuses on adding features such as graphics to increase motivation with the understanding that adding too much may be a distracter. The focused-more-is-more approach provides students ample time to learn course material while using added features such as graphics and challenging learning situations. Multimedia options provide students with an opportunity for multiple attempts at mastering content in a lesson without the need of the instructor repeating the same material (Miller, 2014). Furthermore, student engagement is enhanced with the use of more than plain text. Hathaway (2013) encouraged the use of multiple types of learning tools to engage students in an online classroom. To increase the benefit of multimedia tools in the online classroom, technology should be introduced early in the class (Niess & Gillow-Wiles, 2013).

Content and instructional strategies. Focusing on principles of andragogy, content in online courses should be learner centered (Chametzky, 2014; Luyt, 2013). Collaboration with peers is another strategy to enhance learning and engagement in online courses (Niess & Gillow-Wiles, 2013). Content should include collaborative activities which have corresponding rubrics detailing criteria for interaction and engagement. The best practices recommended for developing content in an online course are a combination of collaborative activities, reflective activities, clear assessment criteria, and integration of technology (Niess & Gillow-Wiles, 2013). Applying course redesign strategies is another effective way for instructors to appropriately transition their face-to-face courses to online courses and successfully integrate technology into the online courses. Furthermore, course redesign reduces cost and raises retention rates (Gilroy, 2006; Spiceland, Spiceland, & Schaeffer, 2015). Aligned with principles of andragogy, course redesign strategies support the use of interactive learning in online settings with the use of multimedia tools (Twigg, 2003). The incorporation of online tutorials, automated feedback, small discussion groups, and a supportive learning community to assist students in content mastery are the key components of course redesign (Twigg, 2005).

While the environment is different, overarching best practices for instruction are similar for both online and face-to-face courses. Miller (2014) presented six principles for effective instruction: peer-to-peer interaction, active student engagement in learning, emphasis on practice and student effort, personalization

to the individual student, variety, and emphasis on higher thought processes. Miller reviewed additional strategies to consider when teaching online courses that are not factors in face-to-face courses. Instructors must consider how to incorporate the use of synchronous activities as online courses usually follow an asynchronous model. Assumptions cannot be made regarding students' ability to use technology in a course. In accordance with the recommendation from Niess and Gillow-Wiles (2013), training must be provided on how to use technology at the beginning of the course. As discussed in the section earlier, multimedia should be included to avoid only learning from text readings.

Considerations for content development. The trajectory of the course should be laid out clearly for the students, and content should be presented in meaningful sections throughout the course (Allen, Kiser, & Owens, 2013). Clarity of assignment instructions is necessary for improved student understanding along with additional time for students to complete online collaborative learning activities (Allen et al., 2013; Miller, 2014). Student learning outcomes, objectives, and assignments must be aligned. As in a face-to-face course, assessment, both formative and summative, is important in an online course (Miller, 2014). Although formative assessment has historically been less emphasized in online courses, it is not only a way to support learning but a strategy to keep students engaged in the online environment (Gikandi, Morrow, & Davis, 2011).

Issues Related to Instructors

The review of literature revealed that issues related to instructors may be summarized into the four specific categories of changing faculty roles, transition from face-to-face to online, faculty time management, and teaching styles as shown in Table 3.

Changes in faculty role. One of the major challenges with online education is the changing role of the instructor (Berge & Collins, 1996; Coppola, Hiltz, & Rotter, 2001; Syverson & Slatin, 2010). Four different roles are identified for online instructors: pedagogical, social, managerial, and technical responsibilities (Berge, 1998). Pedagogical roles refer to teaching methods; social roles are the way instructors establish social relationships with the students; managerial roles include administrative and organizational tasks; and technical responsibilities are the technical support that instructors provide for students. Educational purposes of online teaching are mainly fulfilled through performing pedagogical tasks (Doll, 1993; Robertson, 2000). Other literature supports the shift in the instructor's role by placing emphasis on the ability to deliver content, transfer from teacher-centered to student-centered education, better communicate, and use technology (Berge & Collins, 1996; Choi & Park, 2006; Coppola et al., 2001; Doll, 1993; Fein & Logan, 2003; Juan, Steegman, Huertas, Martinez, & Simosa, 2011;

Table 3. Instructors' Issues and Related Sources.

Instructors' issues	Sources
Changing role of faculty	• Berge (1995) • Berge and Collins (1996) • Choi and Park (2006) • Coppola et al. (2001) • Doll (1993) • Fein and Logan (2003) • Juan et al. (2011) • Neely et al. (2010) • Robertson (2000) • Syverson and Slatin (2010) • Yang and Cornelliuss (2005)
Transition from face-to-face to online	• Anderson et al. (2011) • Baran, Correia, and Thompson (2011) • Chang, Shen, and Liu (2014) • Coppola et al. (2001) • Crawley et al. (2009) • Edge et al. (2000) • Fein and Logan (2003) • Juan, et al. (2011) • Limperos, et al. (2015) • Osika et al. (2009) • Sherry (1996) • Sundar (2007; 2008)
• Communication barriers • Student vs. faculty lead courses • Instructor's interest • Teacher preparation programs	
Time	• Anderson et al. (2011) • Capra (2011) • Cavanaugh (2005) • Fein and Logan (2003) • Juan et al. (2011) • Li and Irby (2008) • Kyei-Blankson and Keengwe (2011) • Moreland and Saleh (2007) • Mbuva (2014) • Tunks (2012)
Teaching styles	• Anderson et al. (2011) • Bawane and Spector (2009) • Choi and Park (2006) • Chang et al. (2014) • Crawley et al. (2009) • Fein and Logan (2003) • Graham et al. (2001) • Juan et al. (2011) • Limperos et al. (2015) • Moreland et al. (2007) • Sundar (2007) • Tunks (2012)

Neely & Tucker, 2010; Robertson, 2000; Syverson & Slatin, 2010; Yang & Cornelli, 2005). The shift from faculty-centered education (faculty lectures) to a more student-centered approach (students decide how they wish to learn) produced a new role for the instructor as a facilitator. The faculty role changes from “performer” in front of the face-to face class in the lecture-style delivery of information to “guides” who must make adjustments to the delivery of content based on their online environments (Coppola et al., 2001). In other words, the online instructor role changes from knowledge transmission agent to a specialist to guide students’ learning process (Juan et al., 2011). In this role, instructors facilitate student learning, rather than teach students (lead lecture). Furthermore, the faculty is available to provide feedback (pedagogical), point to the tools (managerial or administrative), and facilitate engagement between students (social relationships; Berge, 1998; Yang & Cornelli, 2005).

The variation in roles is made more challenging when the instructor is a novice teacher in the online settings. According to Choi and Park (2006), novice instructors find online courses involve a heavy workload, technology issues, and student–teacher interaction. The major pedagogical challenge stems from the inability of instructors to seamlessly transfer their face-to-face course materials to the online environment (Choi & Park, 2006). Furthermore, Fein and Logan (2003) explained that faculty face challenges at three phases with online education: the design, the delivery, and the follow-up. In the design phase, faculty must take into consideration how students learn and what to include to assure that the class materials interest and engage students. Some of the suggested resources are the use of media, lecture notes, and other sources that can add to the class materials. Challenges with delivery are that many faculty members are not able to translate the materials into the online medium. The follow-up phase deals with storage of the information, access to it later, and the dissemination of materials. In addition to the misunderstanding about the transfer of materials from one medium to another, faculty may feel a disconnection between the design produced by the curriculum and design team, and the actual delivery of class content (Neely & Tucker, 2010).

Transitioning from face-to-face to online. The challenge to effectively transfer what is taught in the face-to-face classroom to online continues to be a problem. Anderson, Imdieke, and Standerford, 2011 stated that they saw one of the main challenges as the “disconnect between the way teachers were taught to teach” (p. 4), and how the course content must be delivered in an effective online classroom. This disconnect, while not new, does present a problem as many of the teacher education programs may not have yet caught up to the evolving online teaching environments. Another challenge outlined by Anderson et al. (2011) is the almost non-existence of institutional expectations for their online courses. These include expectations of teachers, students, courses, and staff. Without clear guidelines and expectations for faculty members to follow,

there is no way to assess the effectiveness of these online courses. Further Anderson et al. (2011) explained that the feedback they received from students seldom helped them in adjusting their teaching as they would in a face-to-face class.

Additionally, the method of online delivery varies from the traditional face-to-face education (Anderson et al., 2011; Fein & Logan, 2003; Juan et al., 2011) in that student interactions are between student and faculty, student and peers, and student and technology. Of course, this shift in the instructor's role must be supported by the technology and the curriculum developers as illustrated in Fein and Logan (2003). Similarly, Coppola et al. (2001) described the role change for instructors as an opportunity to facilitate interactions between students and their peers. Although the interactions may vary among LMSs, they must take place to assure success of the course. Many instructors struggle with the delivery of the content and engagement of their students due to lack of visual and face-to-face contact with their students (Crawley, Fewell, & Sugar, 2009), thus feeling less control over how to adjust their classes.

Communication barriers. Communication challenges include the effectiveness of faculty communication with language barriers (Sherry, 1996) and communication via various technological modes (Limperos et al., 2015; Sundar, 2007, 2008). The changing role of faculty also impacts the communication between faculty and students. In a face-to-face classroom, instructors take their cues from students' verbal and non-verbal interactions in the classroom (Coppola et al., 2001). It is difficult to take these affective cues when the faculty is unable to see their students' faces (Crawley et al., 2009). In a study of 20 online instructors, Coppola et al. (2001) found that there were not appropriate feedback methods in place and what was in place was similar to courses in the format of earlier distance or correspondent education. In correspondence courses, feedback is provided through regular (snail) mail and usually not as timely as necessary for corrective measures. Online faculty need to be comfortable with the technology and how to use it to be successful. Thus, it becomes necessary to provide sufficient training for faculty on the most current technologies as well as engaging them in the development of online classes. Another group of researchers drew upon mass communication theories and concluded that online lectures delivered via multimodal formats including audio and text created more positive student experiences as compared with lectures offered via only one mode (Limperos et al., 2015).

Student versus faculty focus classes. In accordance with the changing role of the faculty is the focus from face-to-face instructor-focused and led classrooms to the more student-led or student-focused classrooms in the online environment. Consequently, the lecture format of the traditional classroom is less likely to work online as the delivery of the content must be adjusted to meet the demand of an evolving interactive environment (Kember & Kwan, 2000). Other issues

include the balance that must be struck between providing “bells and whistles” of the technology (Fein & Logan, 2003, p. 47) and the content necessary to assess student learning outcomes. As Fein and Logan (2003) suggested, the “bells and whistles” do not necessarily provide the best outcomes for students who might be easily distracted by the videos, graphics, and other such technologies within the class, thus losing the opportunity to engage in the class discussions that might provide learning. These considerations result in the need to expand the same level of resources in the design of the class and technology support (Edge & Loegering, 2000) as maintaining the online presence and teaching the class. As such, instructors often find teaching online courses more time consuming than face-to-face courses.

Instructors' lack of interest in online courses. Many instructors who teach face-to-face are not interested in teaching online (Fein & Logan, 2003; Osika, Johnson, & Buteau, 2009). One of the major issues is that these instructors have been teaching face-to-face for years and do not feel comfortable switching to the online format. This discomfort is the fear of the unknown, or it may be related to the inability to connect with students within the online environment. In accordance with fear of the unknown, many instructors were afraid that they would be replaced by computers (Berge & Collins, 1996; Osika et al., 2009). Some faculty find the online environment cold and distant for students and have not yet made the connections between the content and how best to deliver their lessons online.

Instructors' comfort level with technology as well as their perceptions of the value of online education plays major roles in their willingness to teach online (Fein & Logan, 2003; Osika et al., 2009). While faculty may be comfortable adding technology to enhance their classes, they might not feel that online courses hold the same value as traditional courses (Osika et al., 2009). In addition, faculty who are comfortable teaching in face-to-face settings and enjoy the students' interaction within face-to-face settings rarely feel that online education can offer the level of interaction endemic to faculty-student engagement. Similarly, lack of training for faculty and administrators (Chang, Shen, Liu, 2014; Mbuva, 2014; Osika et al., 2009) is among the reasons for the low-comfort level of instructors to teach online.

Instructor preparation programs and the online medium. The other issue is that instructors of online education preparation programs may not know how to prepare instructors for transition from the traditional face-to-face training to the online teaching (Baran, Correia, & Thompson, 2011). One of the challenges identified in a number of studies is the fact that instructors are bringing their traditional styles of teaching to online and that it does not appear to be working (Coppola et al., 2001). Therefore, the concern is that there should be another way to look at online teaching approaches and online faculty preparation programs considering new strategies to aid in promoting better education for students.

Time. One of the major issues faced by instructors is the demand on their time, as it takes quite a bit of time to prepare, plan, and teach an online class (Capra, 2011; Fein & Logan, 2003; Humphries, 2010). It takes faculty two times as long to prepare and teach online than face-to-face, thus spending more time per student to facilitate the class (Cavanaugh, 2005). In a time comparison study, a faculty teaching an economics class that he has taught both online and face-to-face spent 155 hours to prepare and teach the course online compared with 62 hours face-to-face (Cavanaugh, 2005). Note that the time difference did not vary with class size—in fact, even smaller classes online demand the same amount of time. The impact of time on class development, design, and facilitation may be a deterrent to faculty interested in online courses (Crawley et al., 2009; McKenzie, Mims, Bennett, & Waugh, 2000). Adjusting such expectations is required to successfully teach online courses (Li & Irby, 2008). Providing support and a learning community for instructors is beneficial in improving the online teaching experience (Kyei-Blankson & Keengwe, 2011). While time is a major factor in online instruction issues, other minor aspects, such as instructors' interest in the modality and teacher education programs, might also be areas of concern.

Teaching styles. While earlier resource challenges such as technology, faculty, and staff availability are no longer dire (Crawley et al., 2009), there remains areas to be addressed, such as effective teaching style. In 2001, researchers came up with seven principles for effective online teaching which were adapted from long-standing face-to-face principles. These principles include good faculty–student interactions, setting expectations for interactions, and, the overarching theme of the seven principles, the expectation of interaction both between faculty and students and students and their peers (Graham, Cagiltay, Lim, & Craner, 2001). Effective communication is another component of that interaction as well as timely feedback to students.

Other suggestions for improving online instructors' teaching effectiveness is to use various e-learning methods and strategies, such as dynamic presentations, laboratory tutorials, simulations, conceptual discussions, interaction and collaboration with students to support their activity, exploration, and knowledge development (Juan et al., 2011).

It is necessary that instructors use the tools provided but also consider how best to present the concepts for the best student learning outcomes. The instructor is the single most important factor in determining student success in an online class (Tunks, 2012). The instructor's ability to communicate, form community, and deliver the appropriate lesson effectively makes all the difference in student learning outcomes. It is further suggested that the interaction that takes place between faculty and students plays a major role in the success of online learning. This interaction must be on a human level, meaning establishing a relationship and the ability to connect with students and help them to feel as a part of the class. Using various new interaction software appear to receive

accolades from students because they had “that personal touch” (p.7). A group of researchers stressed the need for faculty to be knowledgeable about the modes of communication available to create creative and successful engagement with students, leveraging the software to produce shared community within the class, and providing students the opportunity to interact in ways that create the ties necessary to be a member of the class community (Bawane & Spector, 2009; Fein & Logan, 2003; Limperos et al., 2015; Sundar, 2007, 2008; Tunks, 2012).

Based on the reviewed studies, one recommendation would be to allow observation opportunities for novice teachers of senior and seasoned faculty to assure that effective practices are transferred to novice instructors. The results of the Anderson et al. (2011) study in which the instructor felt that student feedback could assist them in adjusting course assignments and learning may be another way to address the issues instructors and students face in the online classroom. Students may be provided with opportunities to share what works or does not work for them after each assignment as they are taking the class. Such an approach might assist instructors in making the necessary adjustments as they teach online courses to assure the best experience for their students. However, the problem with this approach is that while the changes might work for the current students, it may not work for future students. The challenge is to assure the instructor’s teaching method has similar effects for the next group of students taking the class. A fine balance must be navigated by online instructors to assure that they are being reflective as well as recognizing the trends occurring in their online class to ascertain the level of change that must take place each time the course is taught.

In addition to having the opportunity to observe seasoned faculty teach online, Choi and Park (2006) suggested that novice instructors might benefit from having specific training on online pedagogical delivery to assure that they understand how students learn and what they will need to do to engage students. Similarly, to assure success of the online environment, instructors should be trained on the use of technology as inconsistent or unreliable technology can become a distraction online and as such negatively impact student success (Fein & Logan, 2003). Struggling with the use of technology may distract faculty and students and negatively affect learning and teaching. Providing students with a back-up plan in case technology fails is important. The support plan may include a technology hotline phone number, email, or chat where students can contact faculty, and a conference line in case the class need to be moved from online to phone conference (Chang et al., 2014; Crawley et al., 2009; Fein & Logan, 2003).

To be effective in the classroom, instructors of online classes must be great listeners and communicators as well as take the time and extra effort to create community and engage students with thought-provoking questions to help move discussions along. Instructors must be able and willing to provide immediate feedback and enforce a safe environment where students feel valued and able to

share their ideas. Staying organized and checking in with students on a daily or weekly basis are important strategies to help build community and keep students engaged in the class. Fein and Logan (2003) suggested FAQs as a resource to assist students in answering their own questions, which provides immediate feedback since students will not need to wait for long to receive answers to their questions. As with all learning environments, it is imperative that the instructors maintain reflectivity to continue to reflect on their work and assure that they can make necessary adjustments for a successful learning experience both for themselves and their students. Overall, recommendations for engagement in the classroom include interactions between faculty and students, students and students (team work, collaborations), and students and content beyond lectures (Crawley et al., 2009; Moreland & Saleh, 2007). While the recommendations provided earlier may not work for all online environments, they offer some guidance for instructors and provide resources that may enhance the role and competencies of online instructors.

Conclusion

A review of literature using Cooper's framework was conducted to identify the issues and challenges related to teaching online courses. Three major categories of findings were identified consisting of issues related to online learners, instructors, and content development. The relationship between the three major classifications and related issues are depicted in Figure 1. Issues related to learners included learners' expectations, readiness, identity, and participation in online courses. Issues related to content included the role of instructors in content development, integration of multimedia in content, role of instructional strategies in content development, and considerations for content development. Issues related to instructors included the four specific categories of changing faculty roles, transition from face-to-face to online, time management, and teaching styles. The results of this review of literature lead to the conclusion that higher education institutions need to provide professional development for instructors, trainings for learners, and technical support for the content development and delivery of online courses to address the challenges in online education and enhance the effectiveness of online teaching and learning.

Higher education institutions play a central role in enhancing the quality of online education by providing support for instructors, learners, and content development. As shown in Figure 1, online education is a dynamic environment whose three major components of instructors, learners, and content continuously affect each other while institutional support also greatly influences the three components.

Online education will be critical for the future of higher education (Allen & Seaman, 2014). Providing a detailed model such as shown in Figure 1 is very valuable, as it shows major issues in online education and informs educators

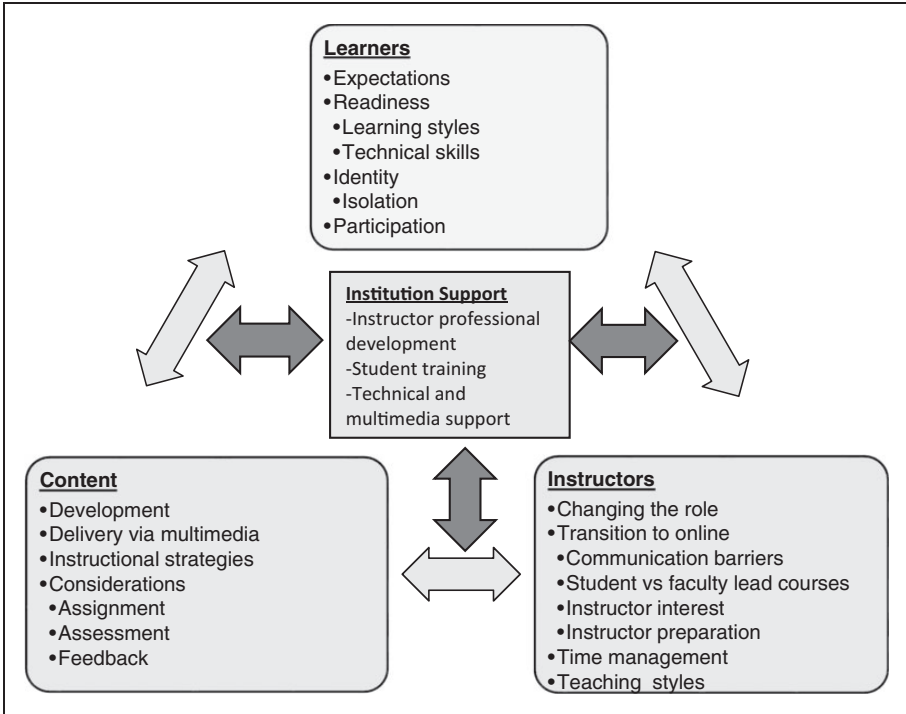


Figure 1. Three major components and the related issues in an online education environment.

about the challenges to be addressed to improve the quality of online education. A number of studies suggested the importance of the support of educational institutions to actively improve the quality of online education (Lion & Stark, 2010; Presteria & Moller, 2001). However, there was a short coming in the literature regarding a classified overview of the issues that need to be supported by the educational institutions. This study bridged the gap and described and delineated major patterns of challenges found in the literature for teaching online courses. It is hoped that higher education institutions consider these challenges and as we recommended, provide professional developments for online instructors, trainings for students, and adequate support for technical issues and multimedia integration to further enhance the quality of online education.

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Author Biographies

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